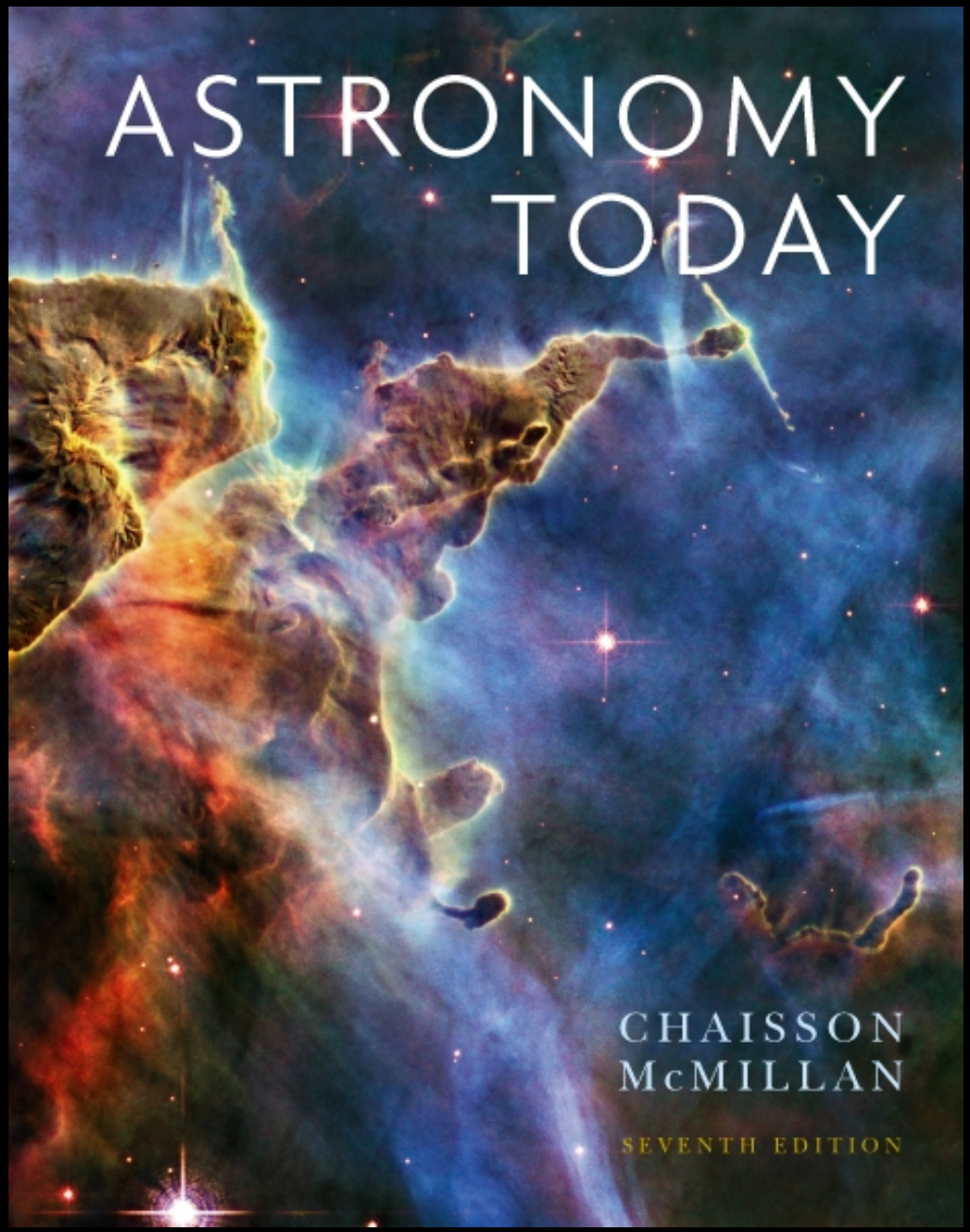


Lecture Outlines

Chapter 6

Astronomy Today

7th Edition

Chaisson/McMillan

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Planetary Astro OUTLINE
Week 12, Day 1

Today:

Turn in Quiz 5
Overview of the solar system (Ch. 6)
- **An inventory**
- **Planet properties**
- **Measuring planet properties**
- **Layout of solar system**

Homework:

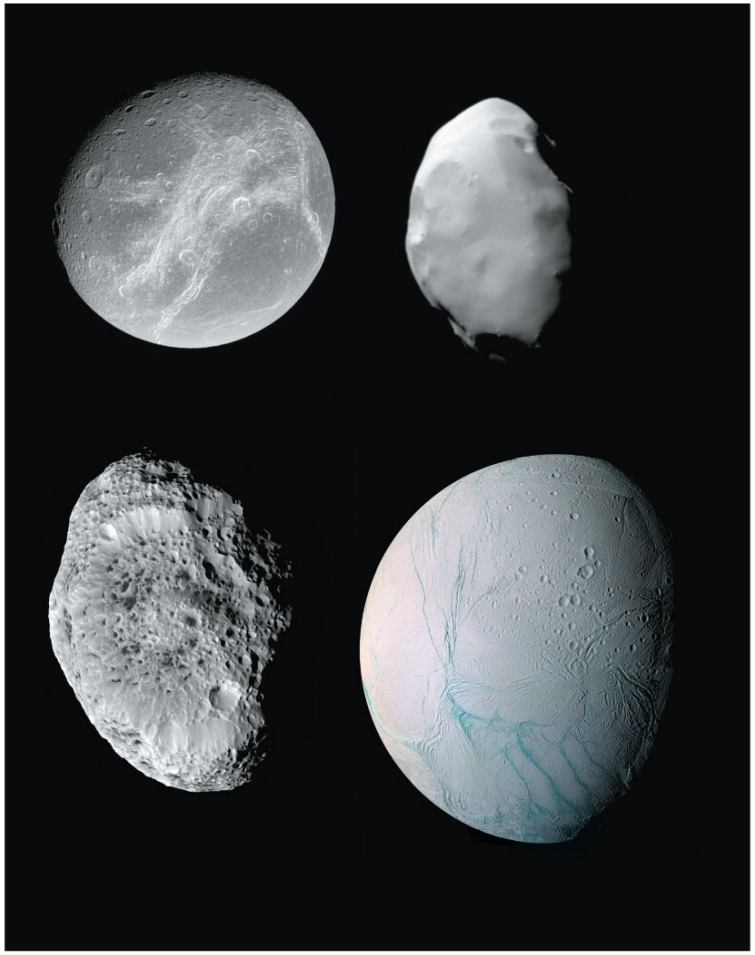
Ch. 6 Disc 1-15 (odd). Mult Choice 1-9 (odd). Due Fri.

Notes:

See new stuff under "NEW STUFF"
Last day to W: this Friday.
Astro News?

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Chapter 6
The Solar System



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Units of Chapter 6

6.1 An Inventory of the Solar System

6.2 Measuring the Planets

6.3 The Overall Layout of the Solar System

6.4 Terrestrial and Jovian Planets

6.5 Interplanetary Matter

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Units of Chapter 6 (cont.)

6.6 Spacecraft Exploration of the Solar System

Gravitational "Slingshots"

6.7 How Did the Solar System Form?

Angular Momentum

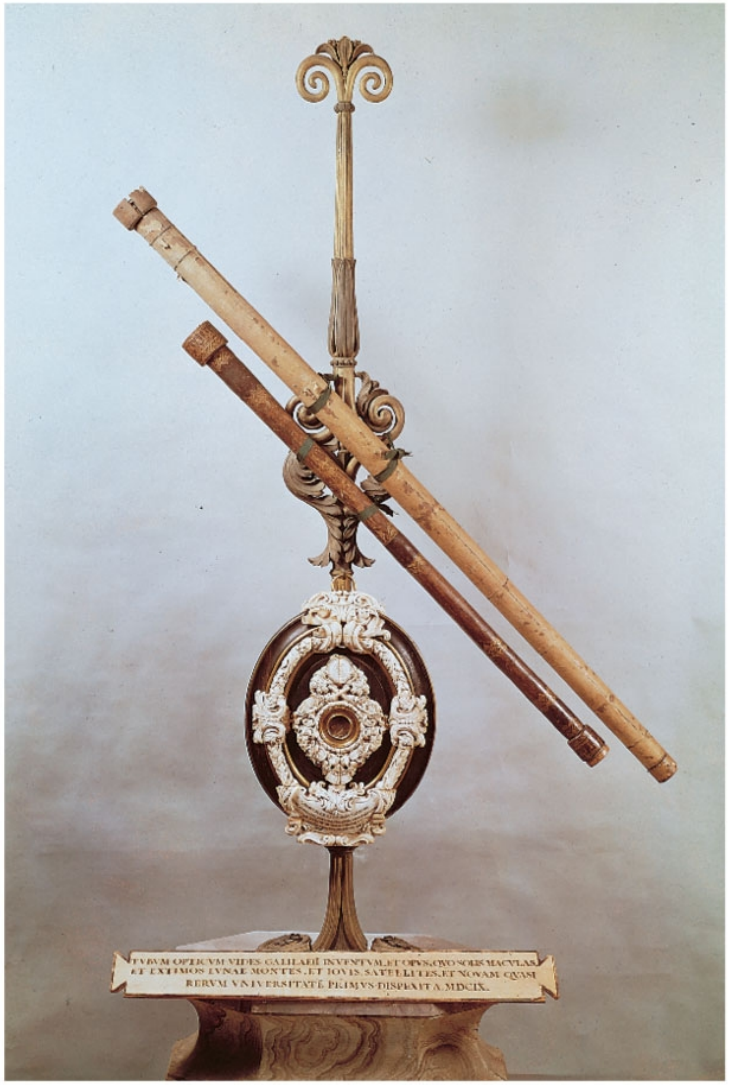
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6.1 An Inventory of the Solar System

Early astronomers knew of the stars, the 7 *luminaries*, comets, and meteors.

The Copernican Revolution gave us the correct ordering and spacing of the planets (in AU).
The telescope led to measuring:

- **the size of AU**
- **stellar parallax**
- **aberration of starlight**
- **and lots of new objects ...**



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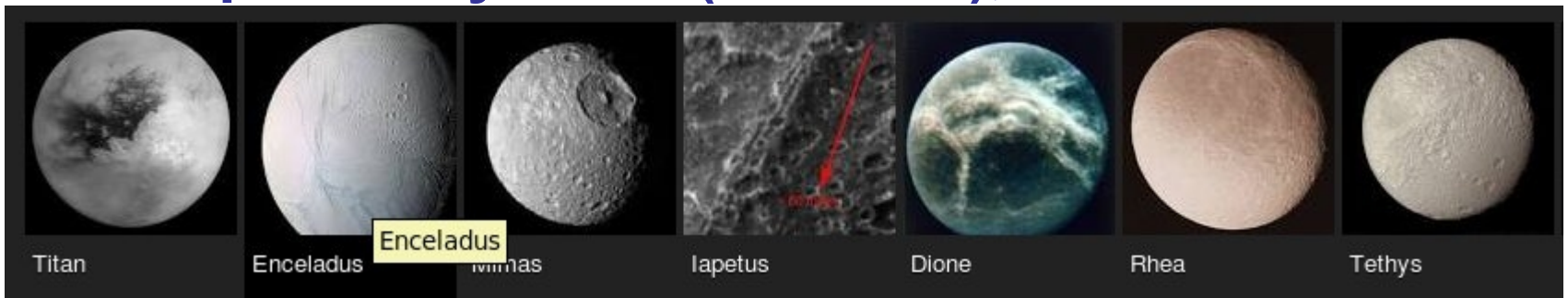
6.1 An Inventory of the Solar System

Now known:

- **Stars: 1**
- **Planets: 8 (added Uranus and Neptune)**
- **Moons: 166 (2011) → 456 (2026, around planets only)**
- **Asteroids: 9 bigger than 300 km, 10^6 over 1 km.**
- **Kuiper Belt Objects: 100 bigger than 300 km**
- **+ many small asteroids, KBOs, comets, and meteoroids (<100m).**

New: 2026 ED138, 600-1500 km

- **"Dwarf Planets: Ceres, 3→ 4 big KBOs, Pluto**
- **Interplanetary: dust (zodiacal), solar wind**



6.2 Measuring the Planets

TABLE 6.1 Properties of Some Solar System Objects

Object	Orbital Semimajor Axis (AU)	Orbital Period (Earth Years)	Mass (Earth Masses)	Radius (Earth Radii)	Number of Known Satellites	Rotation Period * (days)	Average Density (kg/m^3)	Average Density (g/cm^3)
Mercury	0.39	0.24	0.055	0.38	0	59	5400	5.4
Venus	0.72	0.62	0.82	0.95	0	–243	5200	5.2
Earth	1.0	1.0	1.0	1.0	1	1.0	5500	5.5
Moon	—	—	0.012	0.27	—	27.3	3300	3.3
Mars	1.52	1.9	0.11	0.53	2	1.0	3900	3.9
Ceres (asteroid)	2.8	4.7	0.00015	0.073	0	0.38	2700	2.7
Jupiter	5.2	11.9	318	11.2	63 115 (2026)	0.41	1300	1.3
Saturn	9.5	29.4	95	9.5	56 293 (2026)	0.44	700	0.7
Uranus	19.2	84	15	4.0	27 29 (2026)	–0.72	1300	1.3
Neptune	30.1	164	17	3.9	13 16 (2026)	0.67	1600	1.6
Pluto (Kuiper belt object)	39.5	248	0.002	0.2	3 5 (2019)	–6.4	2100	2.1
Hale-Bopp (comet)	180	2400	1.0×10^{-9}	0.004	—	0.47	100	0.1
Sun	—	—	332,000	109	—	25.8	1400	1.4

**A negative rotation period indicates retrograde (backward) rotation relative to the sense in which all planets orbit the Sun.*

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Planetary Astro OUTLINE
Week 12, Day 2

Today:

Return Quiz 5
Overview of the solar system (Ch. 6)
- **Measuring planet properties**
- **Layout of solar system**
- **Terrestrial vs Jovian planets**
- **Interplanetary Matter**

Homework:

Ch. 6 Disc 1-15 (odd). Mult Choice 1-9 (odd). Due Fri.

Notes:

Homework Ch. 16 mean = 9.6/10 (check #2 on luminosity)
Last day to W: this Friday.
Observe Sat 8-9 pm? (Ask on Friday)

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6.2 Measuring the Planets

- **Orbital period** found from synodic period
- **Distance from Sun** from period (Kepler's 3rd law)
- **Radius** known from angular size and distance
- **Masses** from moons and Newton's laws
- **Rotation period** by watching surface features
- **Density** can be calculated knowing radius and mass
- ***Uncompressed density***: the density of a planet if gravity were not compressing it

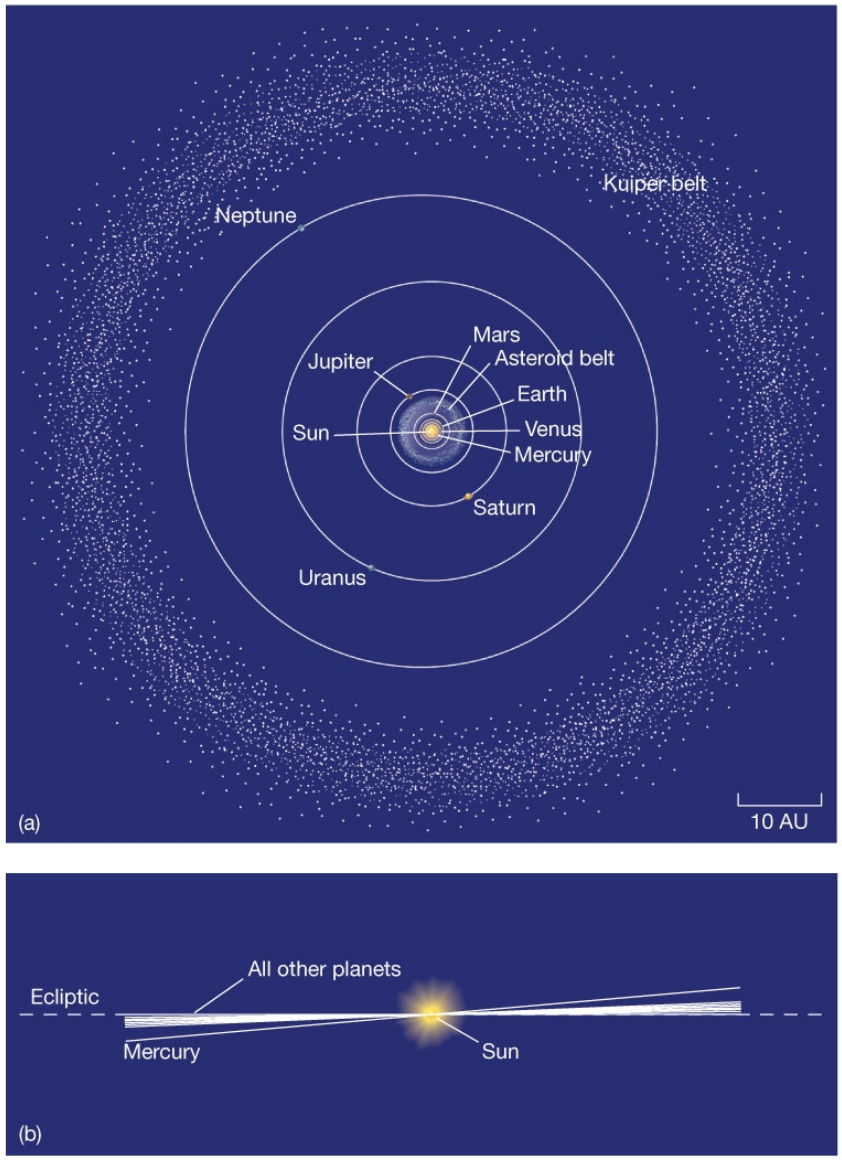
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6.3 The Overall Layout of the Solar System

All orbits but Mercury's are close to the same plane.

All planets and objects orbit in the CCW sense. (Some exceptions exist among the comets.)

Spacings between orbits tend to get increasingly larger.



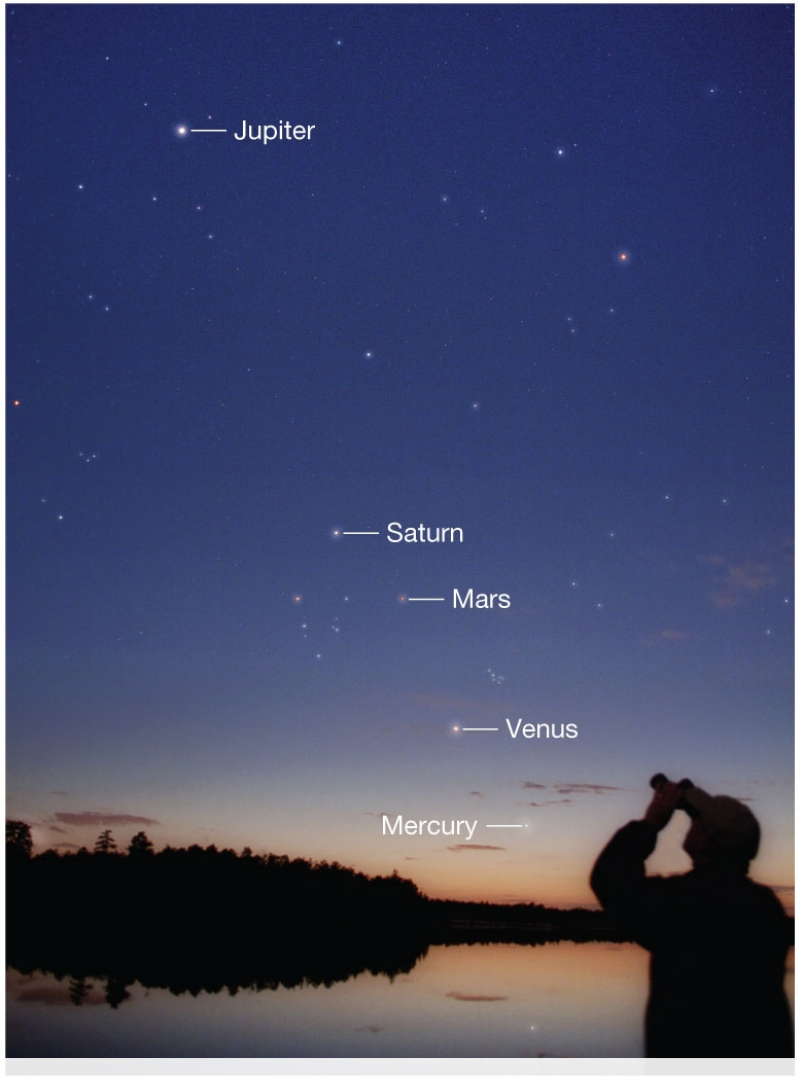
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6.3 The Overall Layout of the Solar System

Because the planet's orbits are close to being in a plane, it is common for them to appear in a straight line as viewed from Earth.

(Photograph was taken in April 2002.)

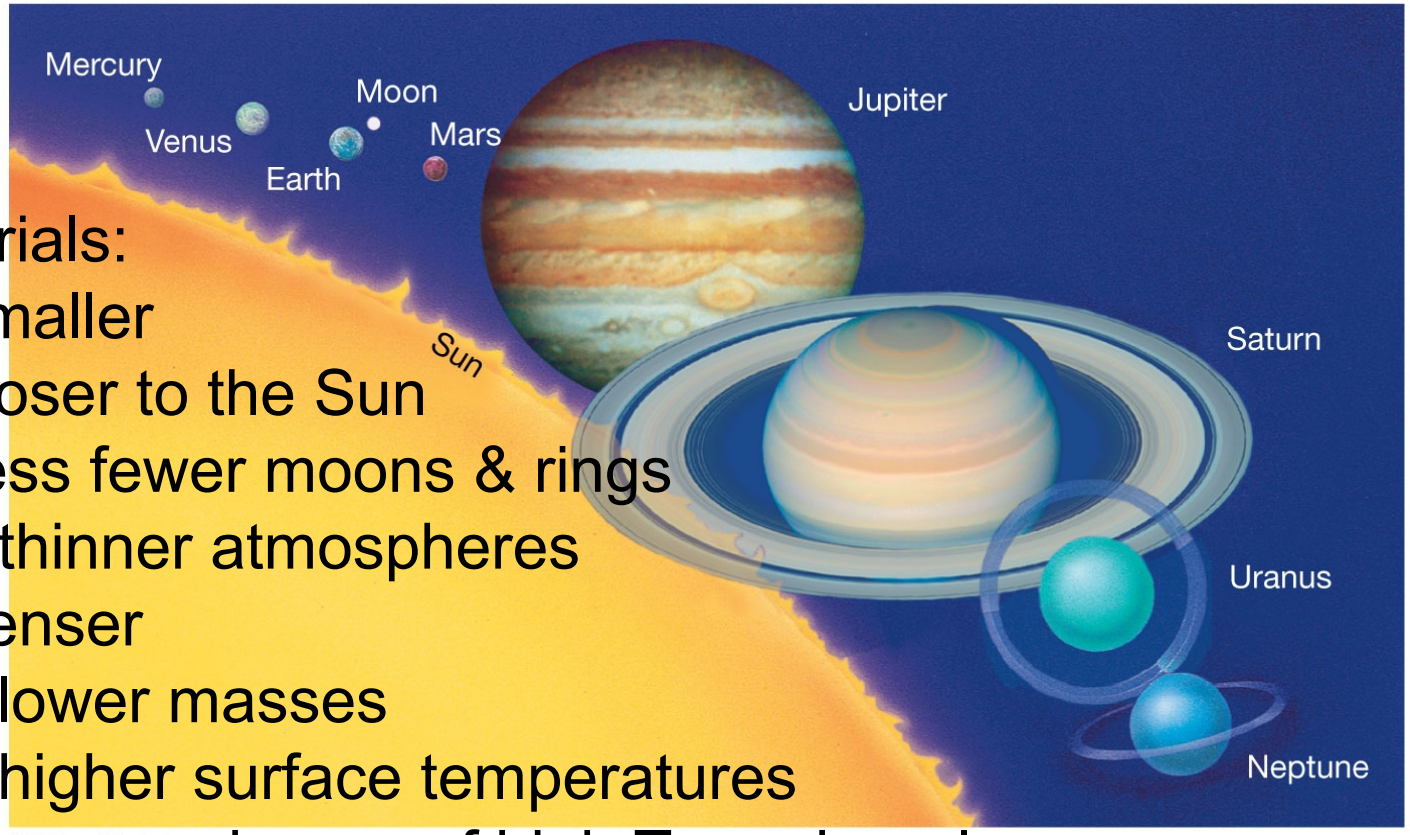


*A 1974 book, "The Jupiter Effect", predicted end-of-world destruction in 1982 due to a similar alignment. Just had one in spring 2022!

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6.4 Terrestrial and Jovian Planets

In this picture of the eight planets and the Sun, some of the differences between the four terrestrial and four Jovian planets are clear.



Terrestrials:

- Are smaller
- Are closer to the Sun
- Possess fewer moons & rings
- Have thinner atmospheres
- Are denser
- Have lower masses
- Have higher surface temperatures
- Are composed more of high T_{melt} minerals

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6.2 Measuring the Planets

TABLE 6.1 Properties of Some Solar System Objects

Object	Orbital Semimajor Axis (AU)	Orbital Period (Earth Years)	Mass (Earth Masses)	Radius (Earth Radii)	Number of Known Satellites	Rotation Period * (days)	Average Density (kg/m^3)	Average Density (g/cm^3)
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Sun	—	—	332,000	109	—	25.8	1400	1.4

**A negative rotation period indicates retrograde (backward) rotation relative to the sense in which all planets orbit the Sun.*

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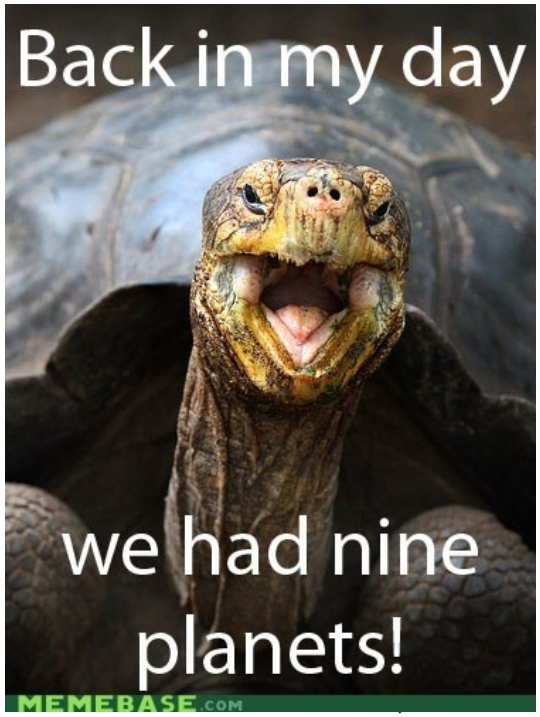
6.4 Terrestrial and Jovian Planets

Pluto was always the oddball.

It is on the inner edge of the Kuiper Belt. *a*=39 AU.

Trans-Neptunian Objects found (Quauar, Makemake, Sedna) of nearly the same size.

Then, Eris was found, even bigger than Pluto and has Moon (Dysnomia).





Planetary Astro OUTLINE
Week 12, Day 3

Today:

~~Return Quiz 5~~ Already did!
Overview of the solar system (Ch. 6)
- Interplanetary Matter
- Spacecraft missions to solar system bodies.
- Formation theories of the solar system

Homework:

Ch. 6 Disc 1-15 (odd). Mult Choice 1-9 (odd). Due today < 2pm

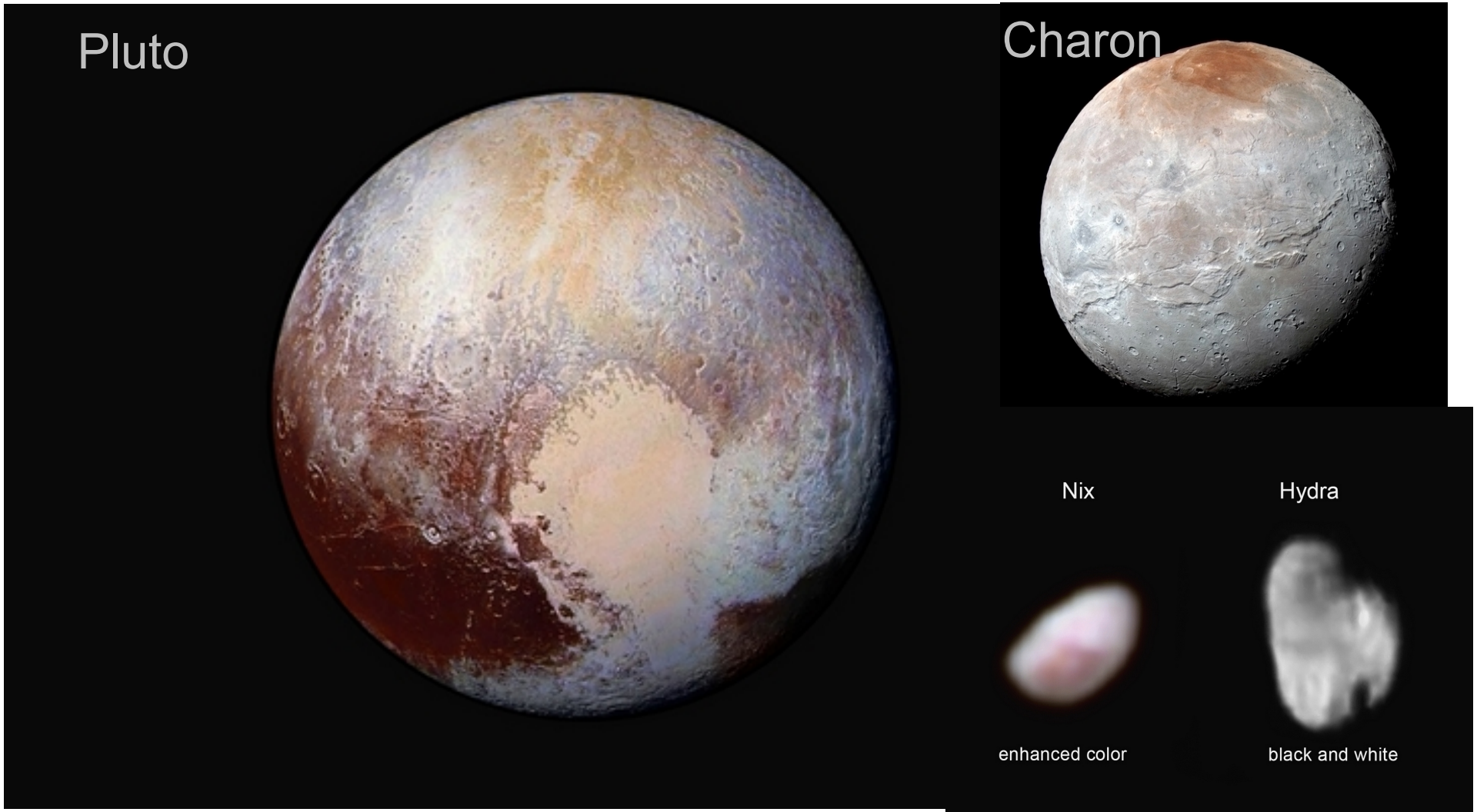
Notes:

Last day to W: is today.
Observe Sat 8-9 pm? – Nope! Looks like Saturday night will be cloudy too!

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6.4 Terrestrial and Jovian Planets

Pluto was visited by New Horizons on July 14, 2015. It is bigger than Eris after all! Has 5 moons!



6.5 Interplanetary Matter

Asteroids!

These asteroids have all been visited by spacecraft.
Samples retrieved from Ryugu and Bennu.



Potentially Hazardous Asteroids (or NEAs):

Amor asteroids approach but do not cross the Earth's orbit from the outside;
Apollo asteroids cross the Earth's orbit, but are typically outside the Earth's orbit;
Aten asteroids cross the Earth's orbit, but are typically inside the Earth's orbit; and
Atira asteroids approach but do not cross the Earth's orbit from the inside.

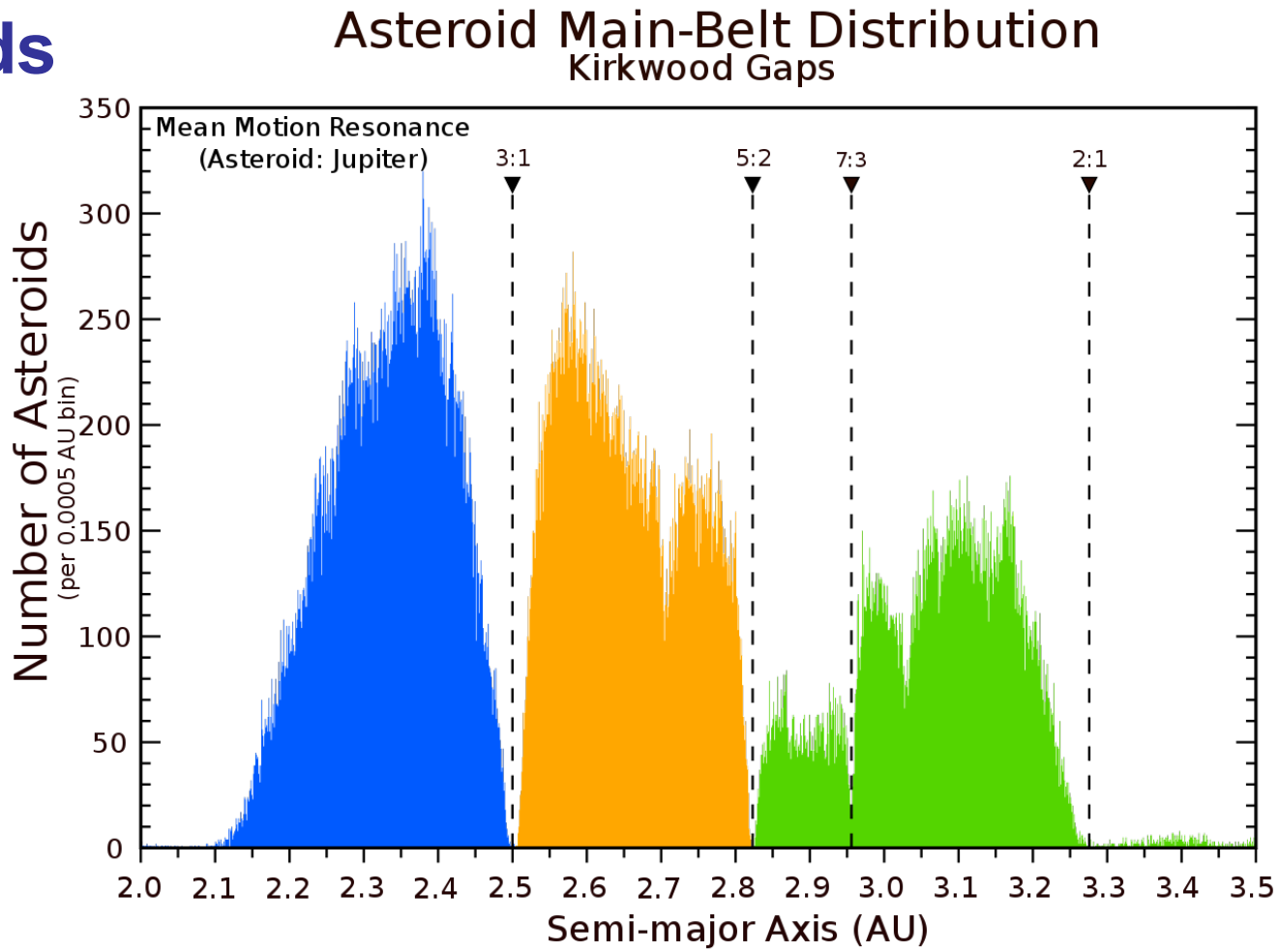
The Torino scale rates hazards from 0-10. Nothing yet higher than a 4 (Apophis, 2004).

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6.5 Interplanetary Matter

Asteroids and meteoroids have rocky composition.
(Meteoroids are smaller than 100 m.)

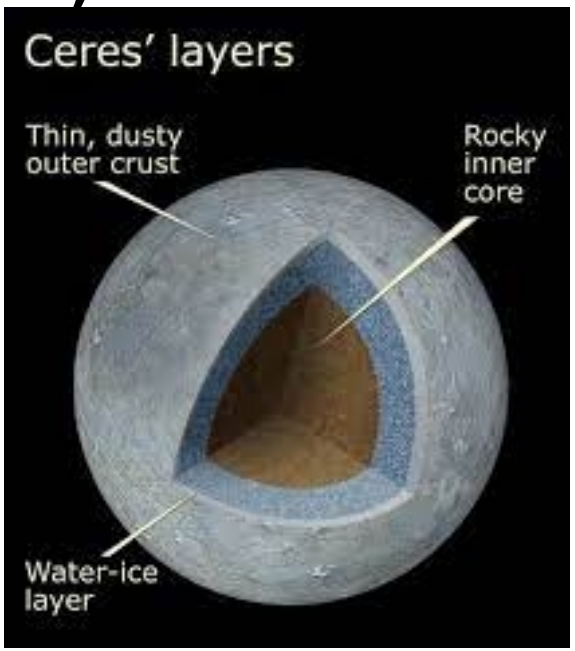
Most asteroids
2<a<3.5 AU.



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6.5 Interplanetary Matter

<u>Asteroids!</u> Biggest asteroid (Ceres†) is spherical and *differentiated*.



Most are potato shaped.

Ex) Asteroid Eros is 34 km long*



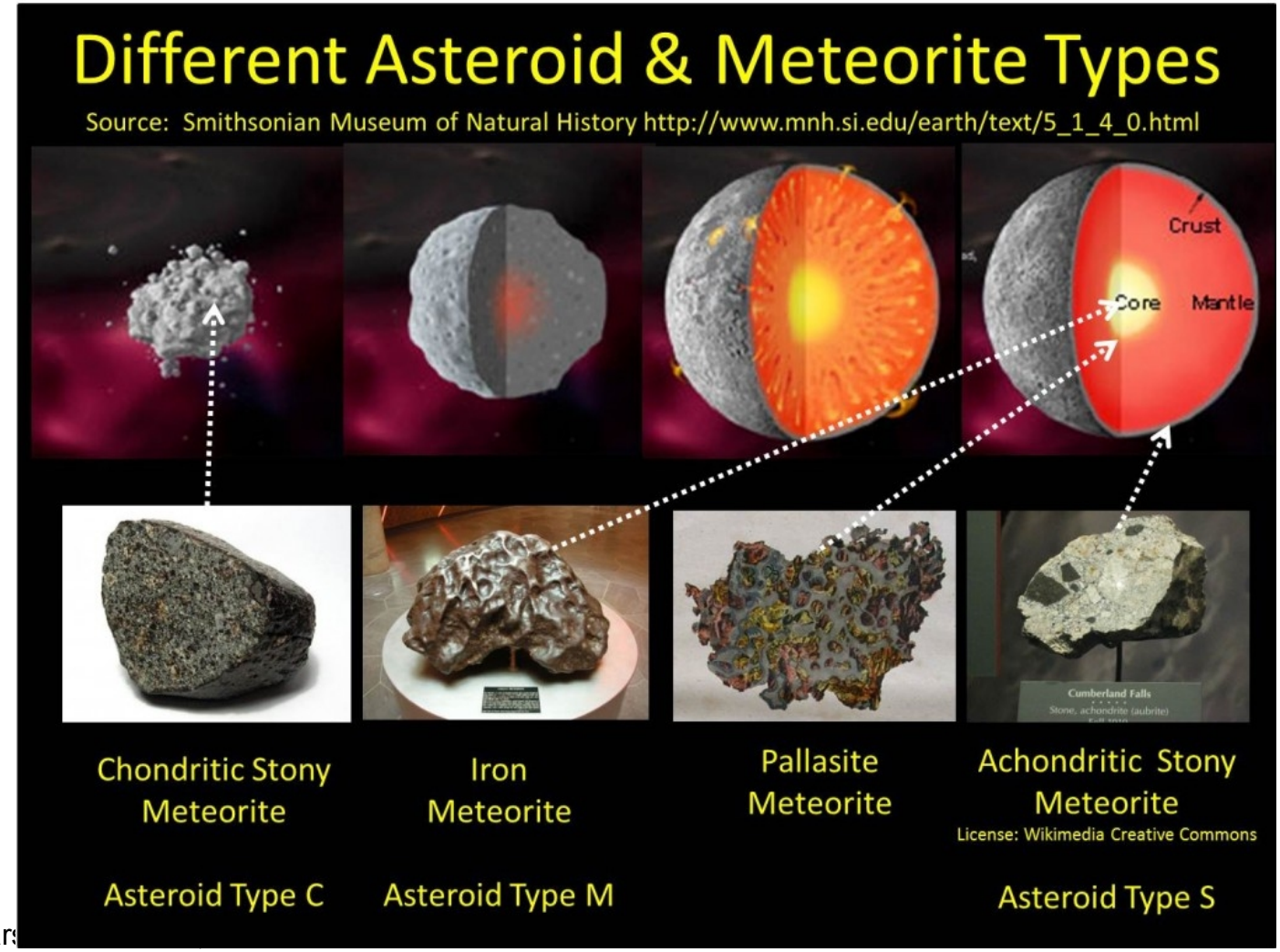
(a)

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†Visited by Dawn in 2015-18. *Visited by NEAR in 2001.

6.5 Interplanetary Matter

Some asteroids are "planetesimals" – building blocks of planets that failed to make a planet.



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6.5 Interplanetary Matter - ***Comets***

Hale-Bopp ‘96

Arend-Roland ‘57



Halley ‘86

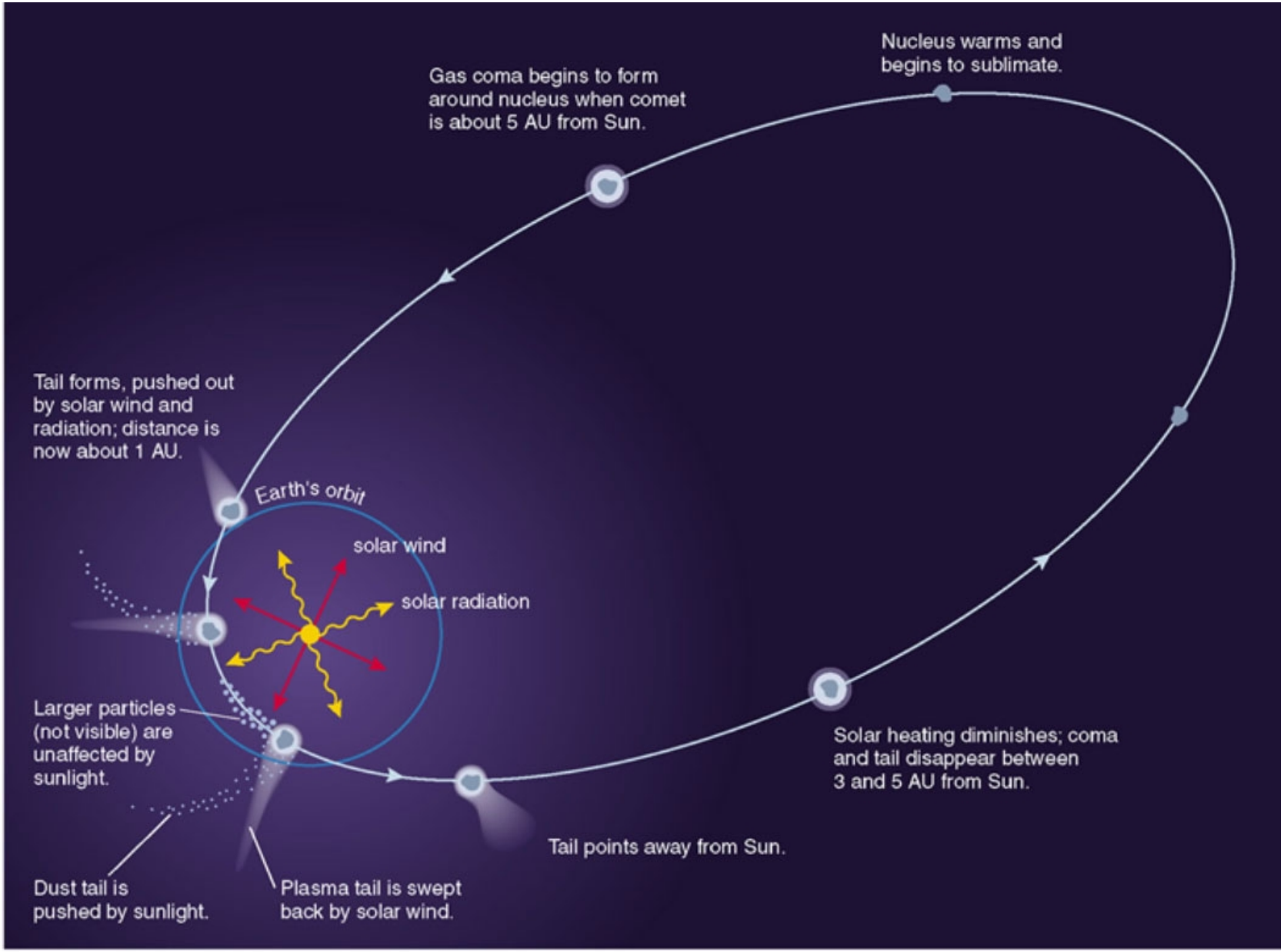
McNaught ‘07

West ‘75

Ikeya Seki ‘65?

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6.5 Interplanetary Matter - ***Comets***



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6.5 Interplanetary Matter

Comets are "dirty snowballs"

Comet Hale-Bopp

Had biggest nucleus (37 mi).

Visible to eye in 1997.

Period ~ 2530 years.

Comet Bernardinelli-Bernstein

Is now the largest known at 60-120 miles. Will be closest (~11 AU) in 2031.



(b)

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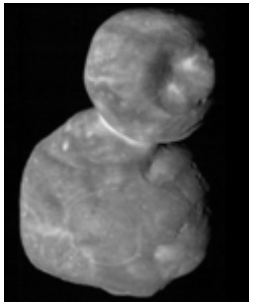
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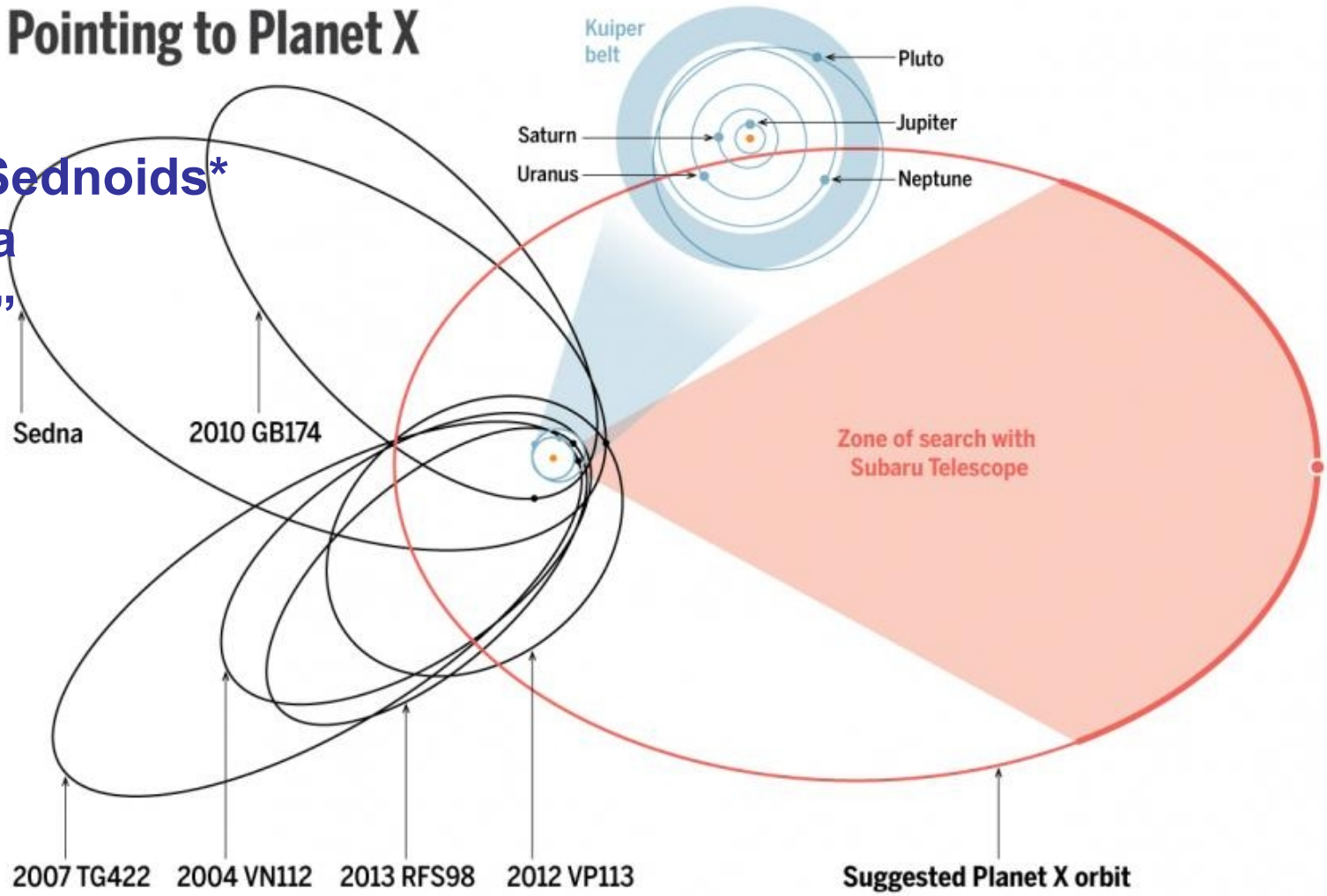
6.5 Interplanetary Matter

Kuiper Belt Objects are beyond Neptune.

Orbits of Sednoids* suggests a "Planet X."

"Arrokoth"





*TNOs with perihelion greater than 75 AU

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6.6 Spacecraft Exploration of the Solar System

Types of Space Missions

***1) Flybys* – the spacecraft flies close to body of interest without orbiting it**

***2)Orbiters* – the spacecraft achieves an orbit around the targeted world**

***3) Probes* – the spacecraft drops down onto target usually gathering data on the atmosphere and surface during the fall.**

By 7Train at the English Wikipedia, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=6780606

6.6 Spacecraft Exploration of the Solar System

Types of Space Missions

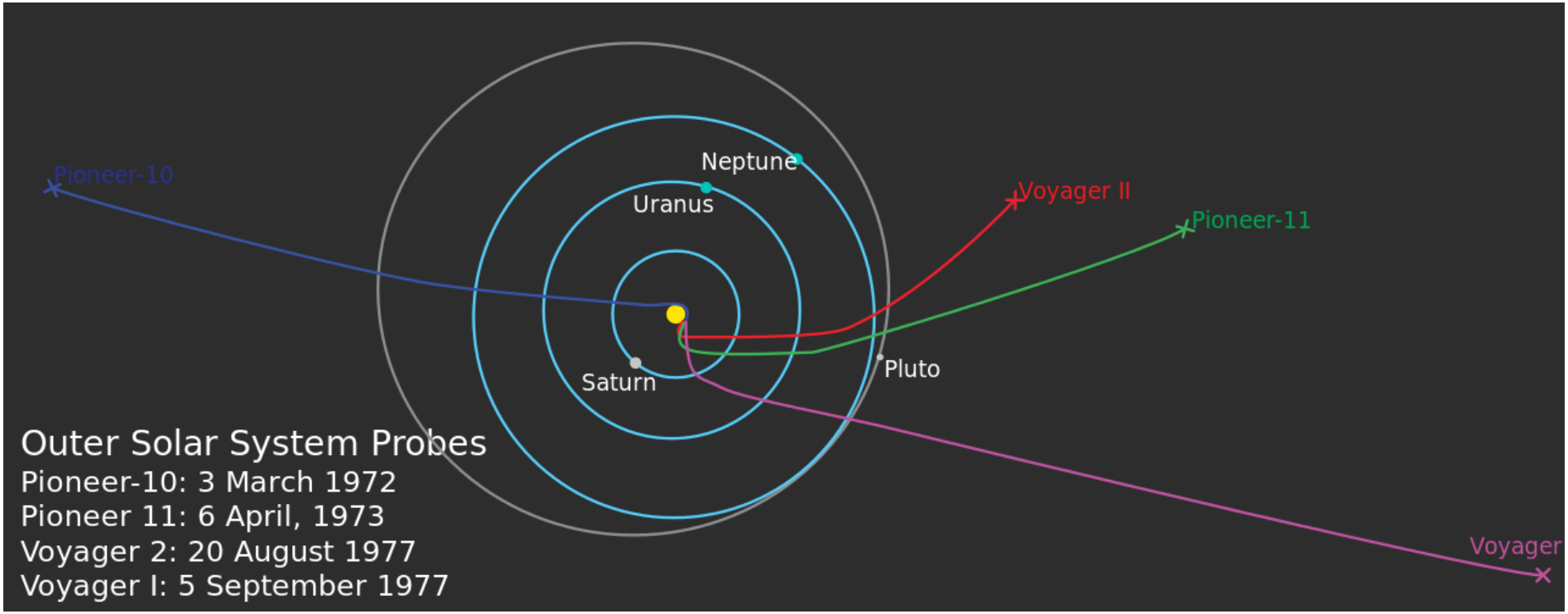
4) Landers **– designed to survive descent and take most of their data from one spot**

5) Rovers **– survive descent and then travel on the surface**

By 7Train at the English Wikipedia, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=6780606

6.6 Spacecraft Exploration of the Solar System

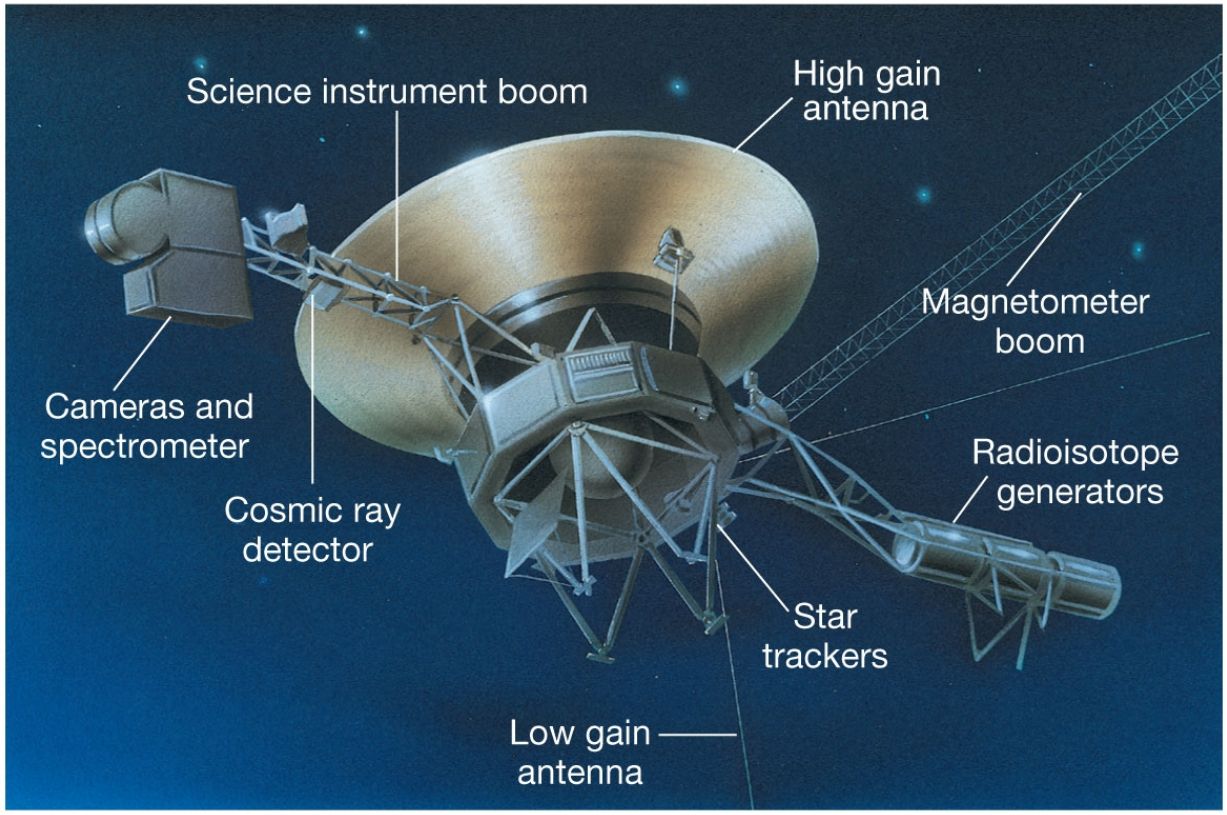
***Pioneer* and *Voyager* had many flyby encounters on their "Grand Tour" missions. All 4 reached escape velocity from the sol. system!**



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6.6 Spacecraft Exploration of the Solar System

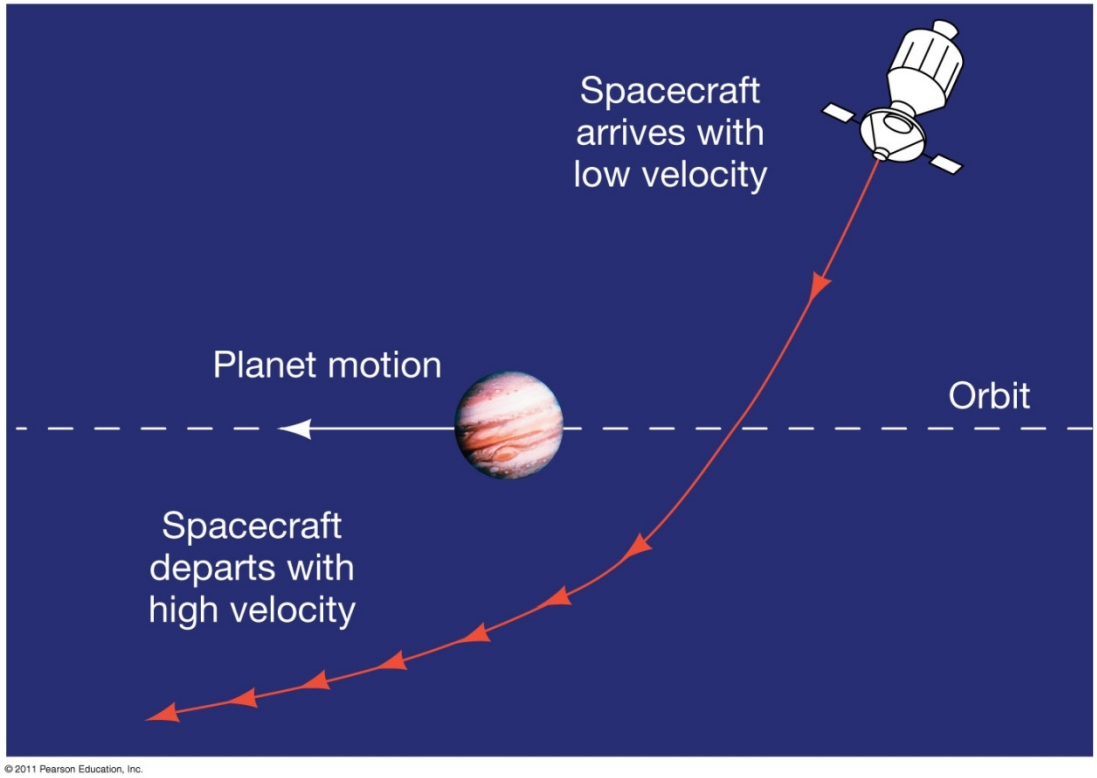
***Pioneer* and *Voyager* flew through the asteroid belt on their way to outer solar system.**
This is *Voyager.*



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Discovery 6-1: Gravitational "Slingshots"

Gravitational "slingshots" can change direction of spacecraft, and also accelerate it



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6.6 Spacecraft Exploration of the Solar System

The most recent Venus expedition from the United States was the *Magellan* orbiter, 1990–1994.

ESAs latest: *Venus Express*, 2005-2015.



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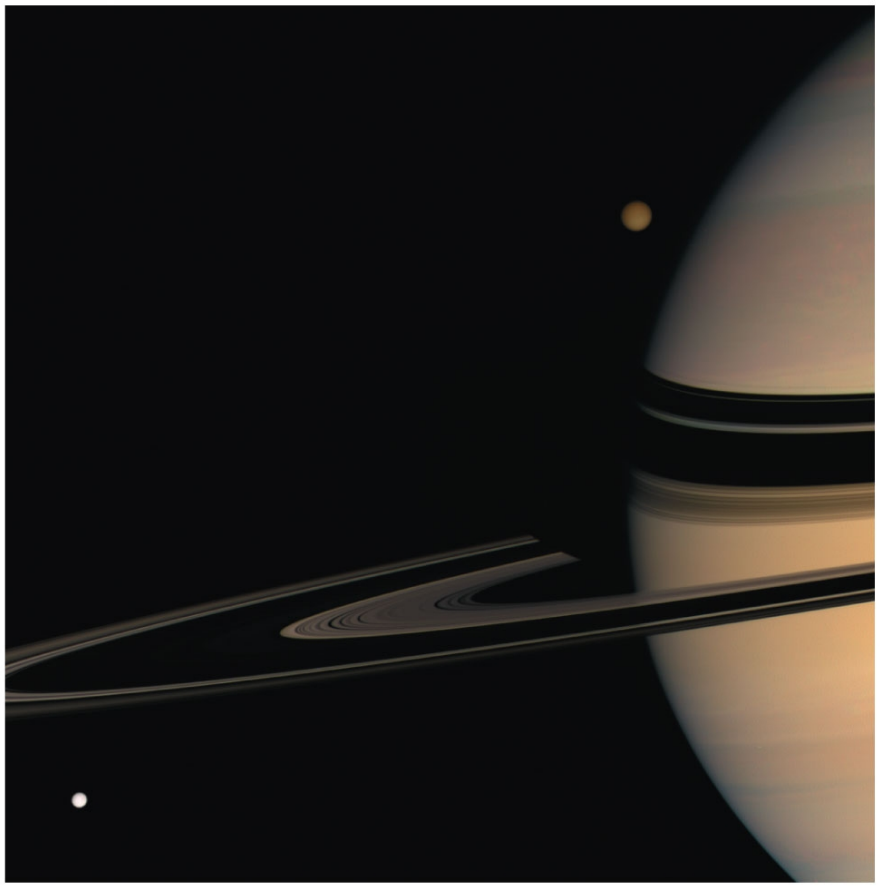
See Globe of Venus

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6.6 Spacecraft Exploration of the Solar System

Cassini-Huygens **mission arrived at Saturn in 2004.**

It has returned many spectacular images while orbiting Saturn.

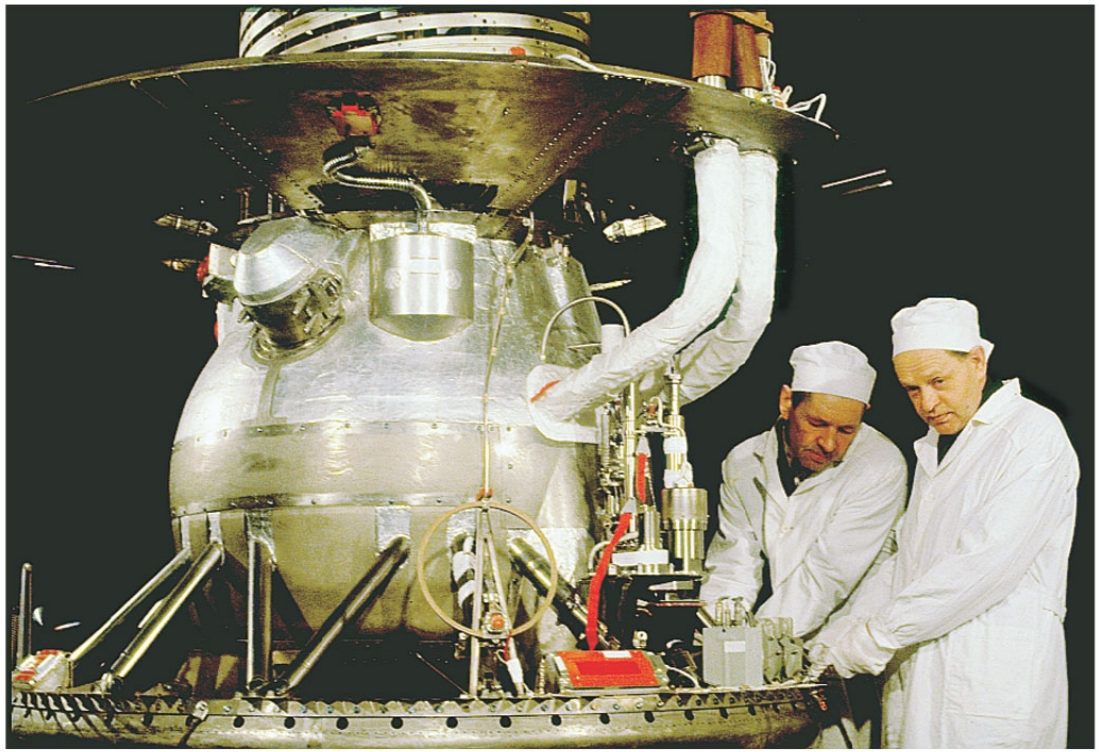


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6.6 Spacecraft Exploration of the Solar System

Soviet *Venera* probes landed on Venus from 1970 to 1978

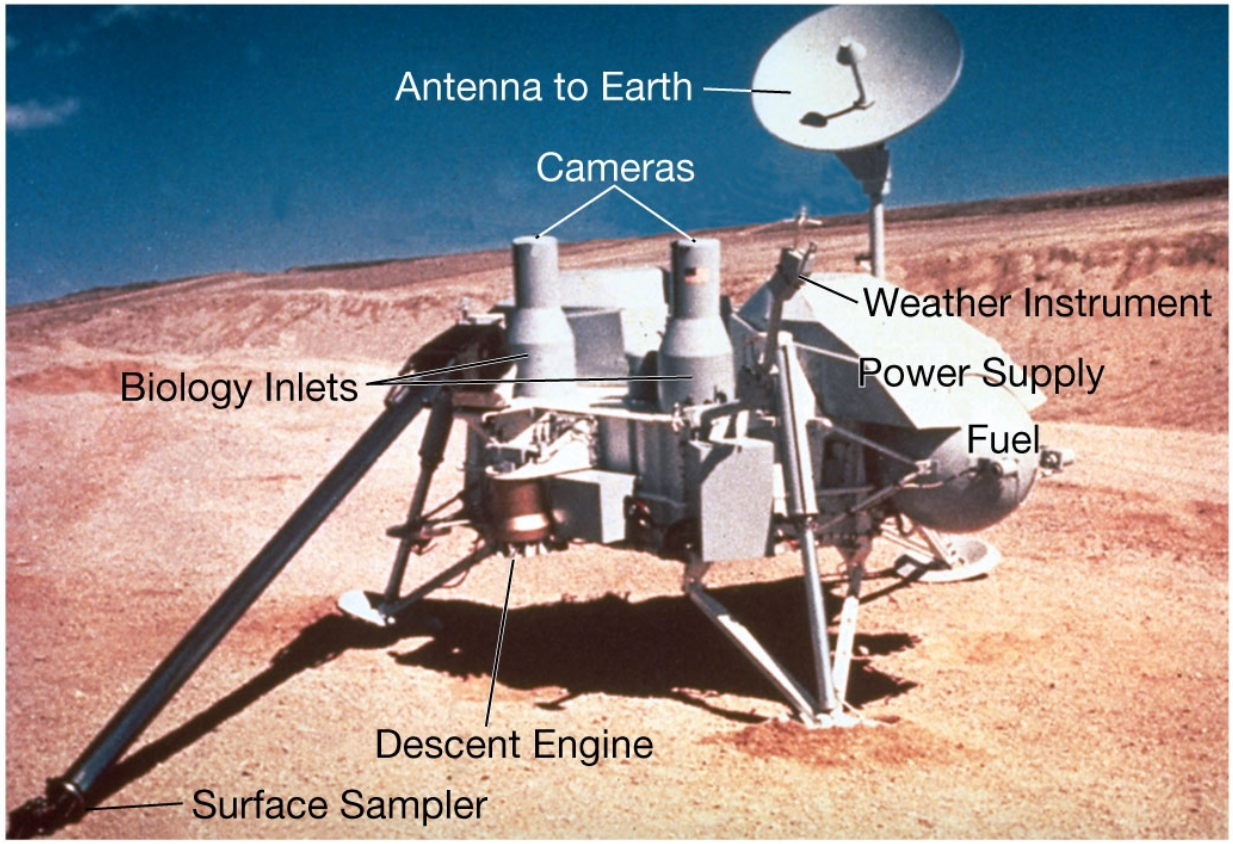


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6.6 Spacecraft Exploration of the Solar System

Viking landers arrived at Mars in 1976

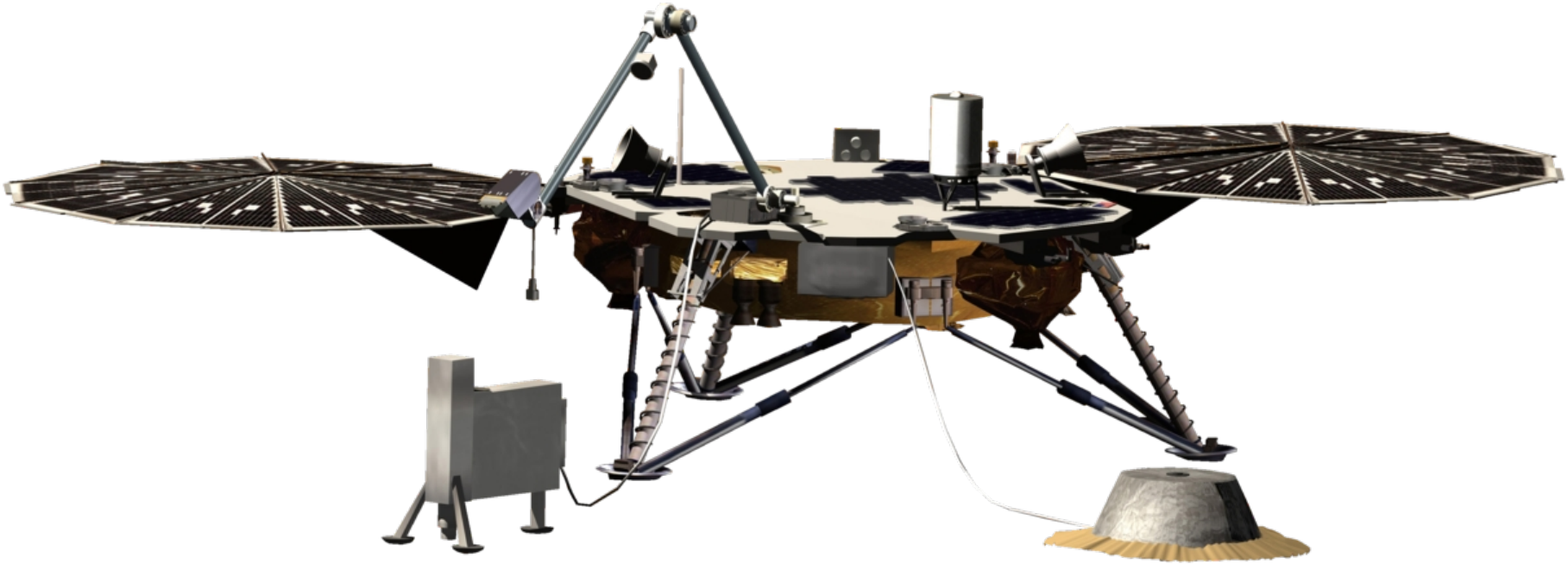


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6.6 Spacecraft Exploration of the Solar System

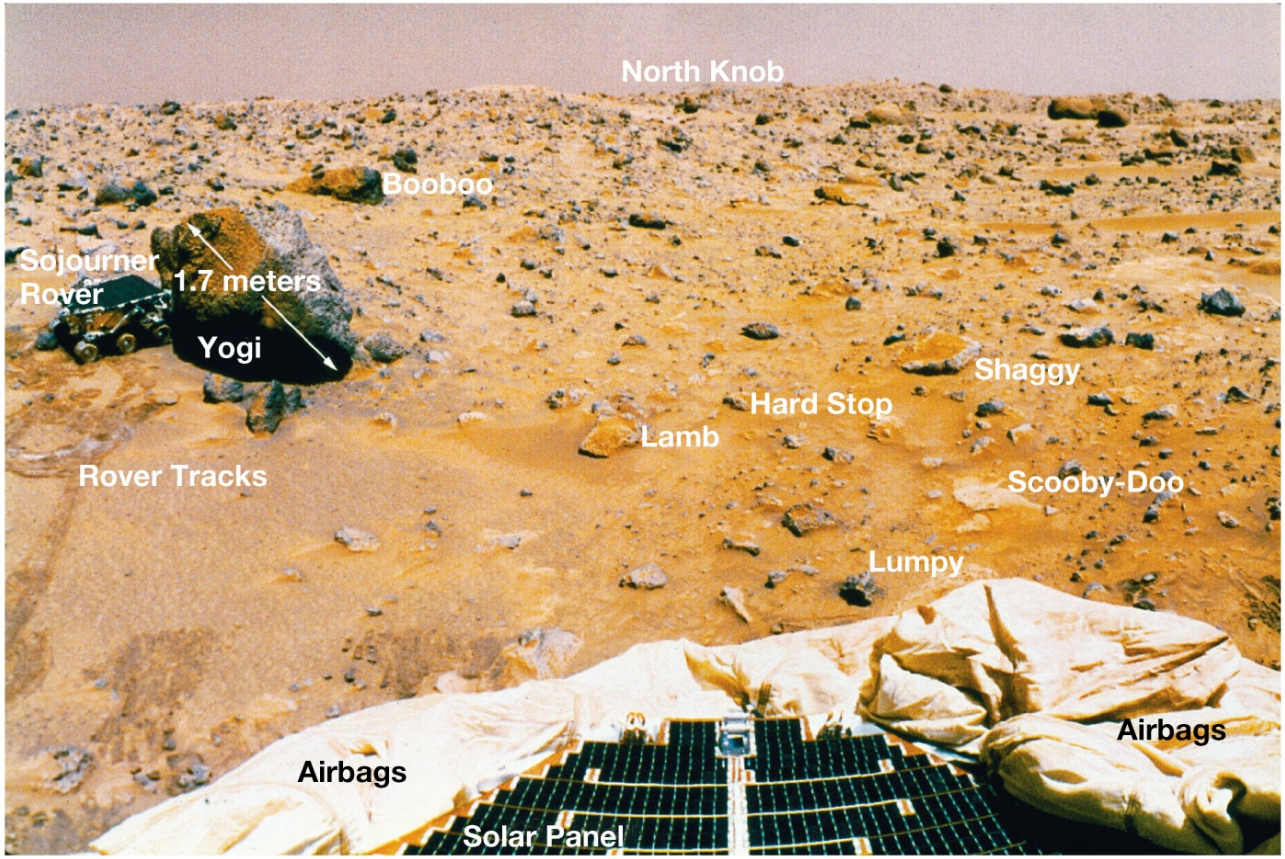
***InSight* lander landed on Mars in Nov. 2018. Has seismometer.**



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6.6 Spacecraft Exploration of the Solar System

Sojourner rover was deployed on Mars in 1997

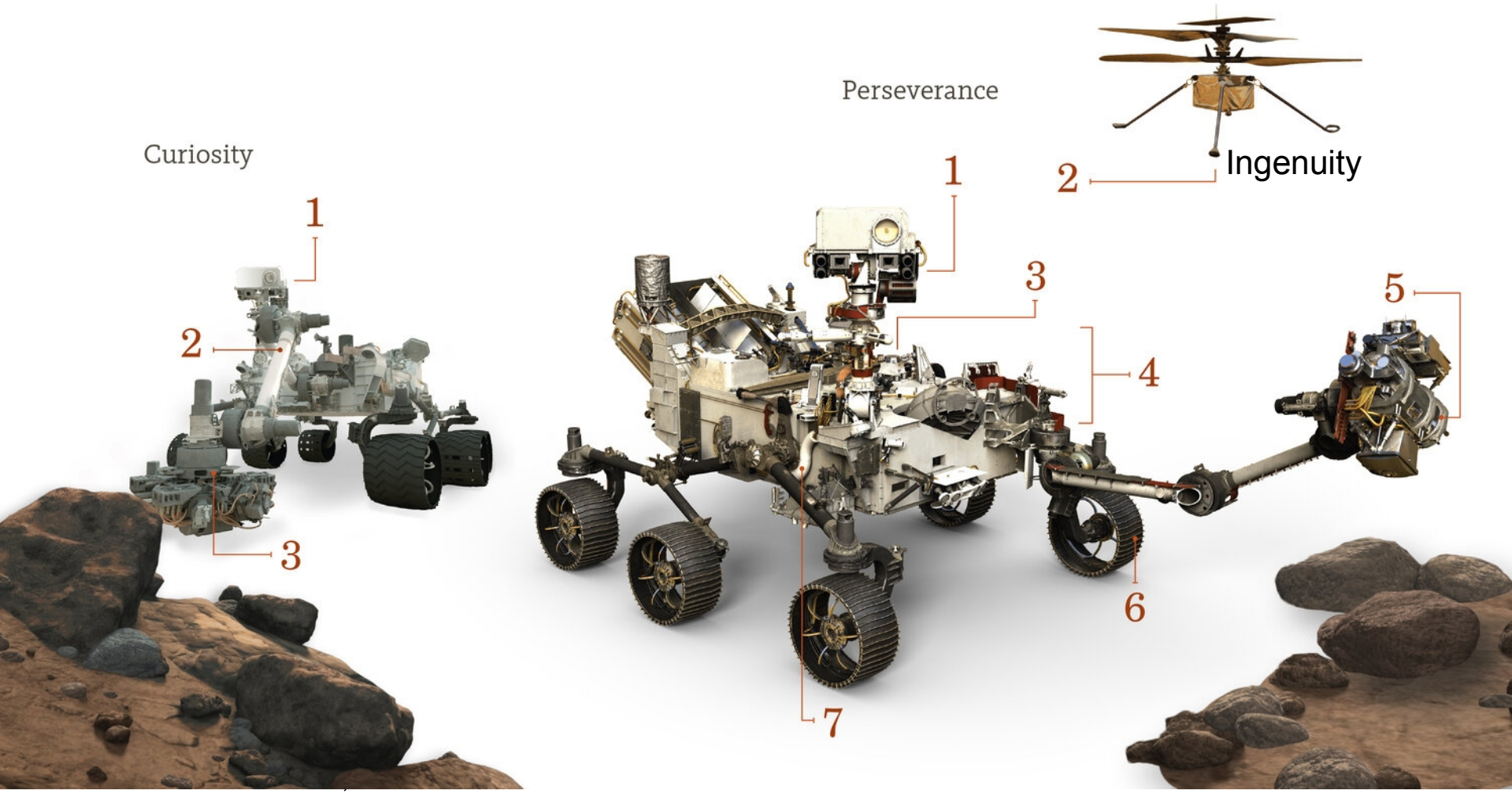


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6.6 Spacecraft Exploration of the Solar System

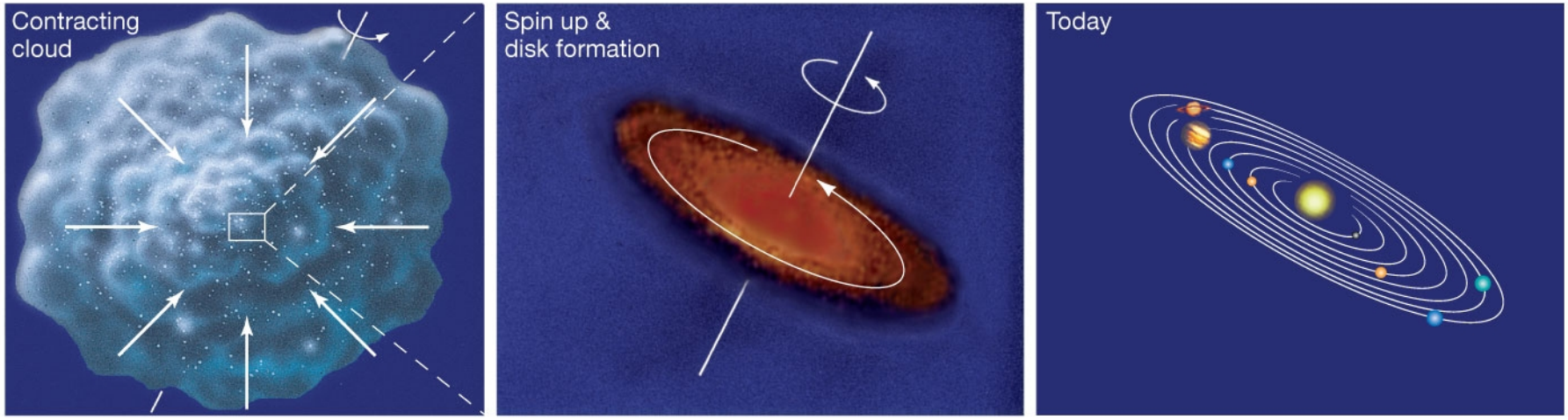
Mars rovers : Spirit, Opportunity, Curiosity, ...



6.7 How Did the Solar System Form?

Nebular theory (Laplace after Descartes):

Cloud of gas and dust contracts due to gravity.
Contracts more easily along spin axis – forms disk.
Conservation of angular momentum means it spins faster and faster as it contracts.



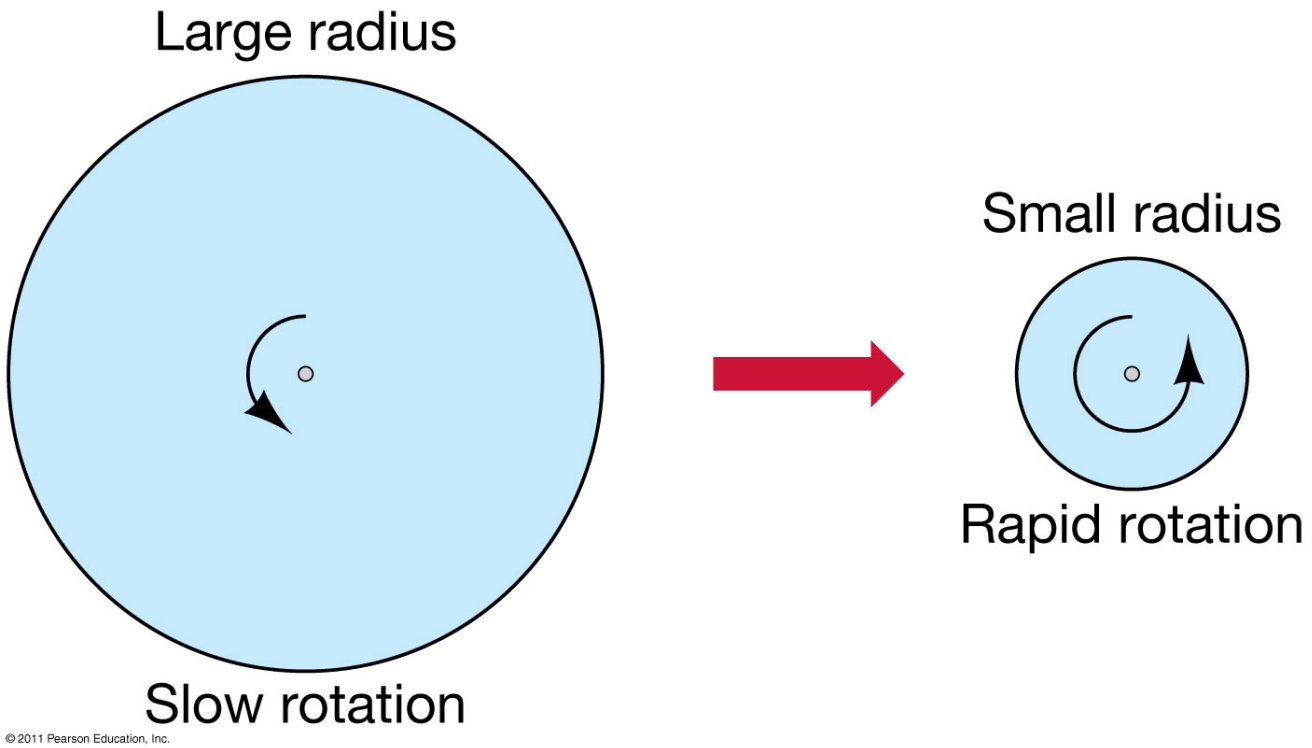
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Planets form out of disk.

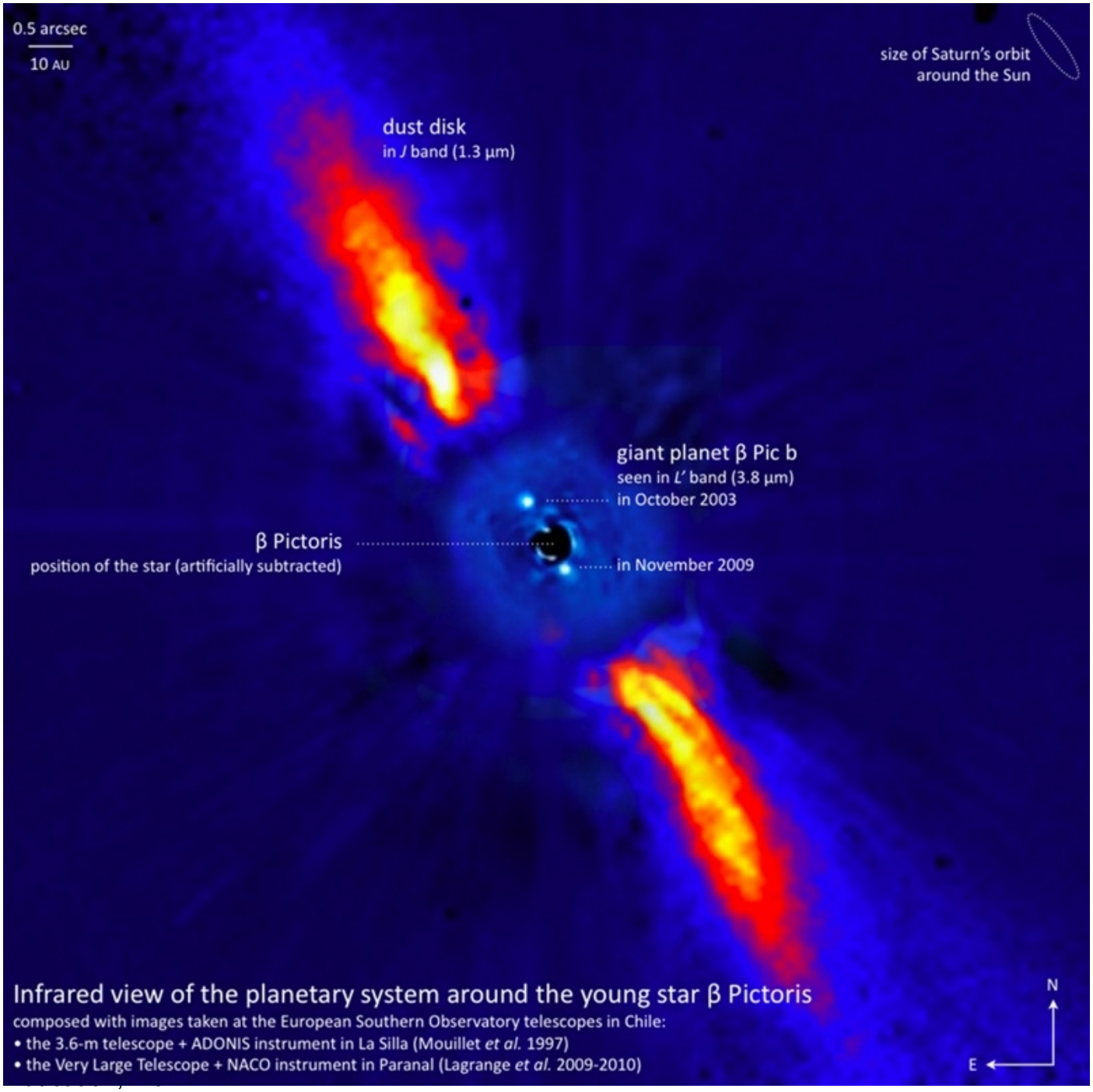
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More Precisely 6-1: Angular Momentum

Conservation of angular momentum says that product of radius and rotation rate must be constant. $L \sim MR^2 \omega$ **(where ω = spin rate)**



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6.7 How Did the Solar System Form?

Dust is important ingredient in the solar nebula.

1) It allows nebula to cool and contract by radiating heat away.

2) It provides nuclei for condensation and accretion onto small chunks.



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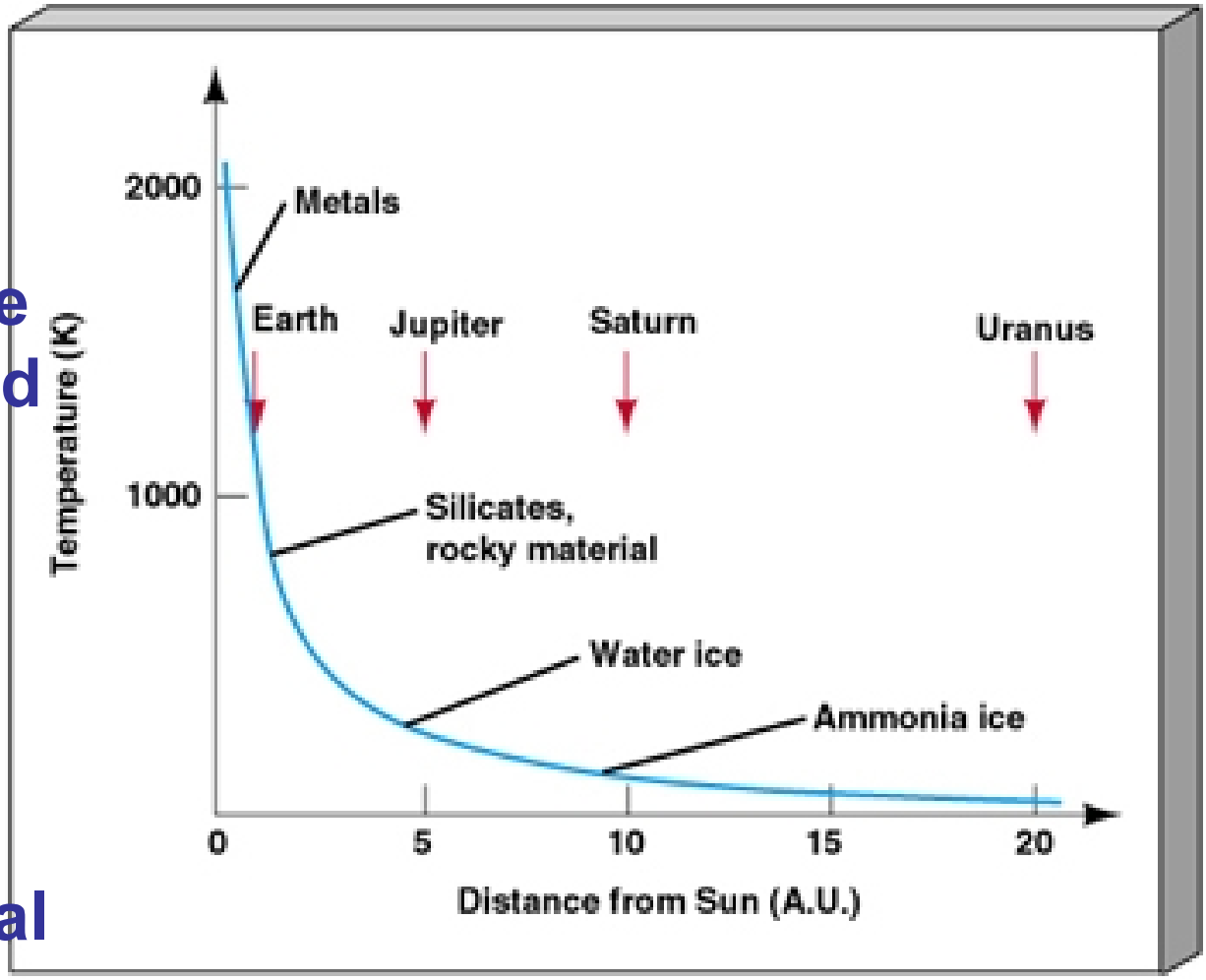
Dust can be seen in inter-stellar clouds as shown here.

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6.7 How Did the Solar System Form?

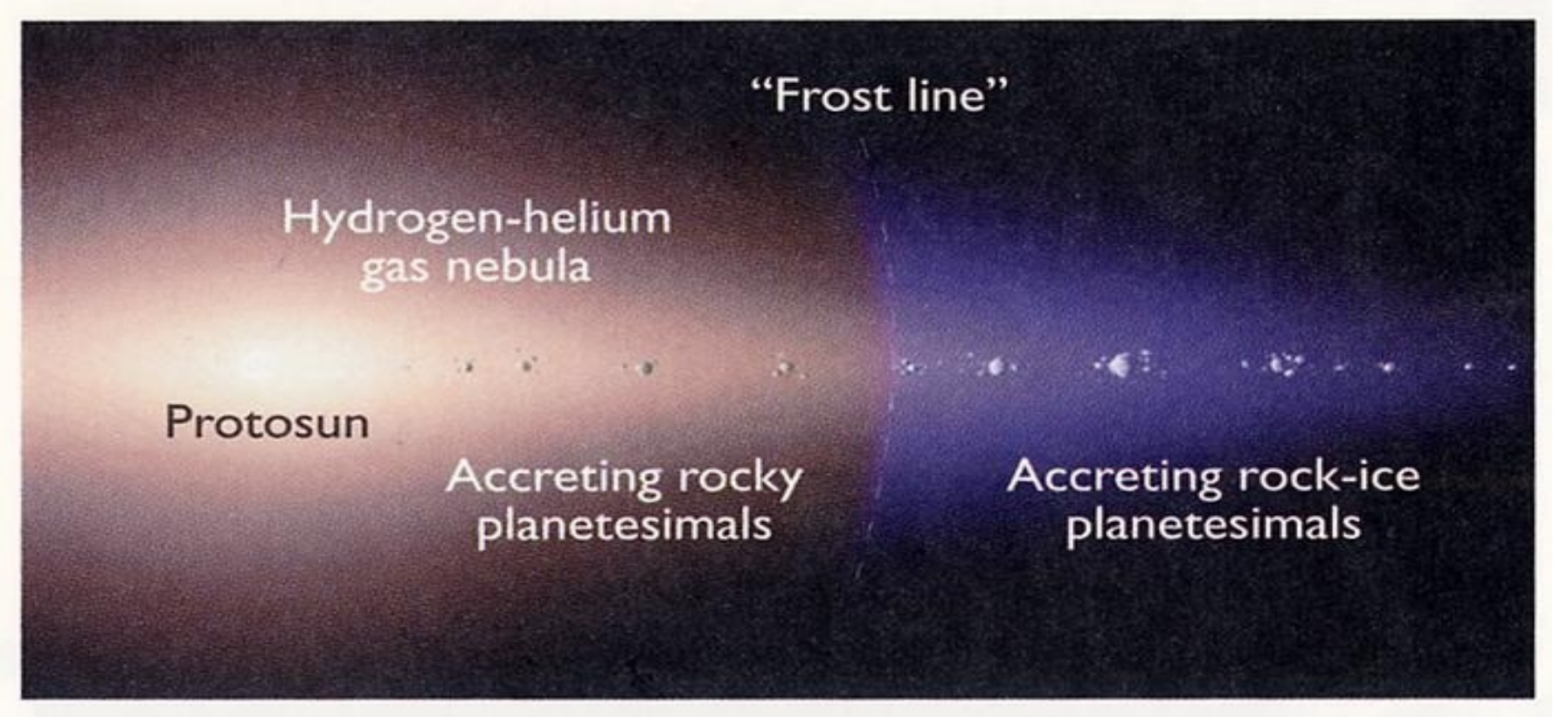
The *condensation sequence* is the way the composition of the dust grains in the solar nebula changed with distance from the Sun.

High melting point materials close to Sun, high and low melting point material far from Sun.



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6.7 How Did the Solar System Form?



Once planets got big enough, they could gravitationally capture H and He gas from solar nebula.

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6.7 How Did the Solar System Form?

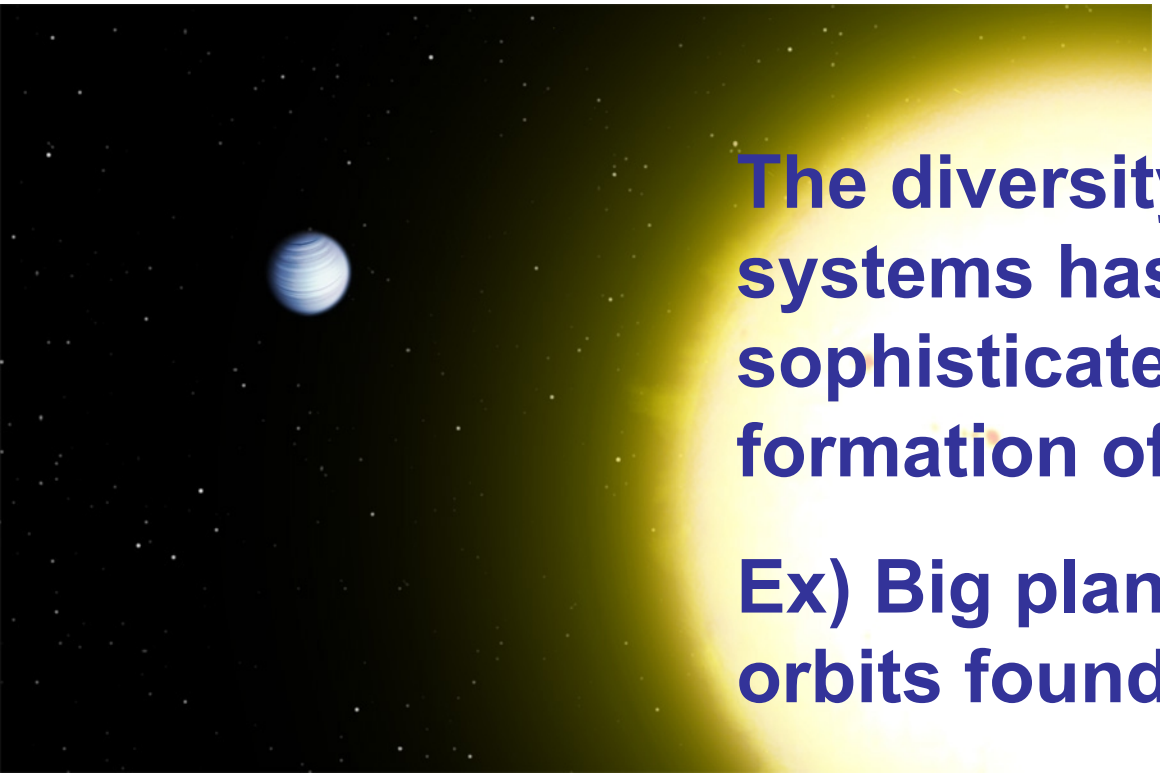
Hierarchical formation terms	**Processes**
Interstellar grains < 1 micrometer	Bonds, Molecular forces
Protoplanetary disk grains >~ 1 micrometer	
Aggregates, fractal aggregates	"Static cling", Stickiness
Pebbles	Slow collisions, Fast collisions
pebble+gas swarms	
Planetesimals	Gravity + collisions + accretion of gas
Protoplanets	

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6.7 How did the Solar System Form?

More than 6400 extrasolar planets have been confirmed! (3337 from Kepler, >1000 from TESS)*

51 Pegasi, the first exoplanet discovered.
Artists conception, from *Wikipedia.*



The diversity of these new systems has required more sophisticated theories for the formation of planet systems.

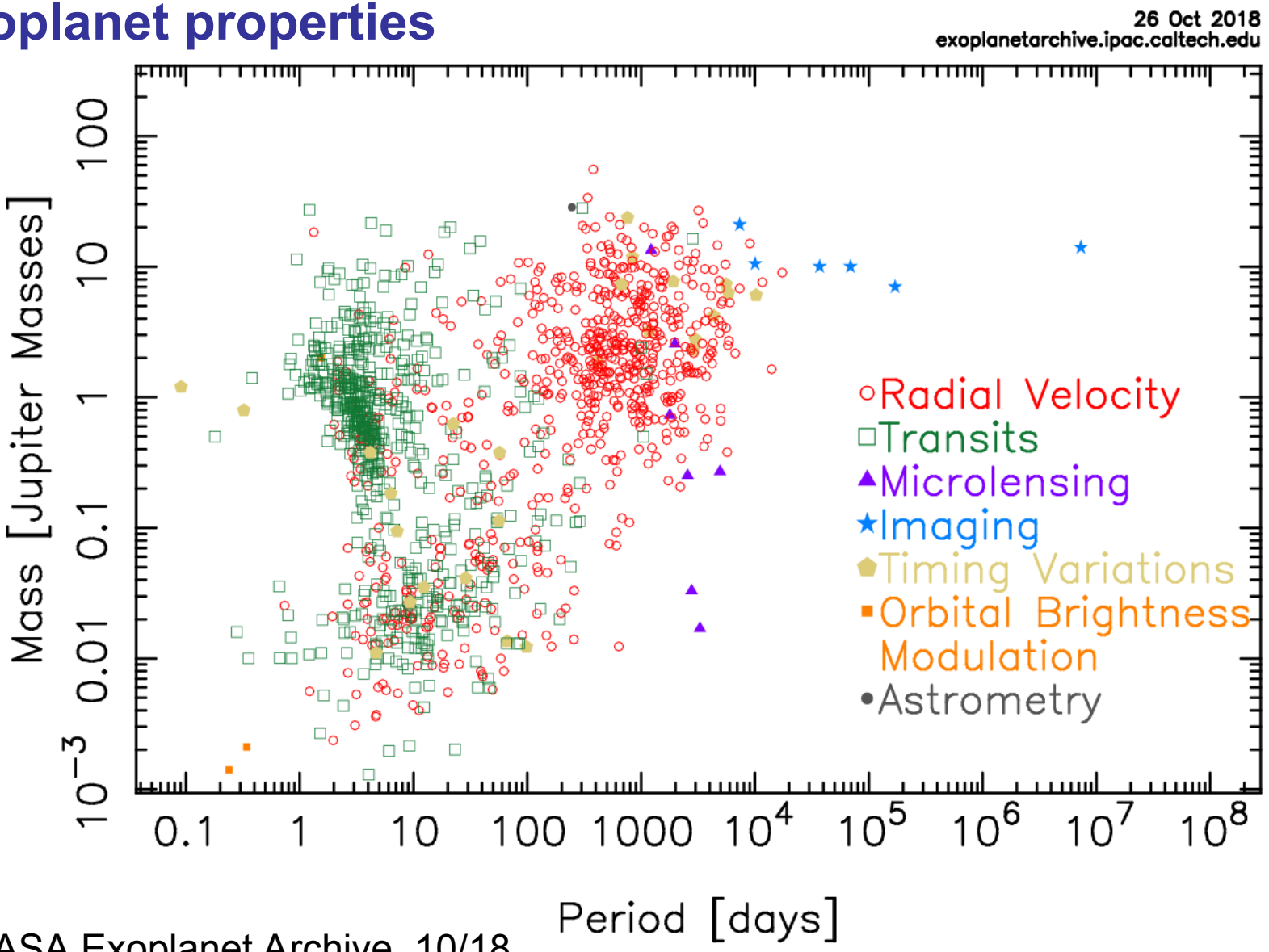
Ex) Big planets with eccentric orbits found.

Ex) Unusually low densities

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6.7 How did the Solar System Form?

Exoplanet properties



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Summary of Chapter 6

- **Solar system consists of Sun and everything orbiting it**
- **Asteroids are rocky, and most orbit between orbits of Mars and Jupiter**
- **Comets are icy and are believed to have formed early in the solar system's life**
- **Major planets orbit Sun in same sense, and all but Venus rotate in that sense as well**
- **Planetary orbits lie almost in the same plane**

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Summary of Chapter 6 (cont.)

- **Four inner planets—terrestrial planets—are rocky, small, and dense**
- **Four outer planets—jovian planets—are gaseous and large**
- **Nebular theory of solar system formation: cloud of gas and dust gradually collapsed under its own gravity, spinning faster as it shrank**
- **Condensation theory says dust grains acted as condensation nuclei, beginning formation of larger objects**

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